Julia Levenshteyn

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EDUCATION

Lake Forest College, Lake Forest, IL **Expected May 2027**
Bachelor of Arts in Chemistry and Data Science, Minor in Mathematics **GPA: 4.0/4.0**

HONORS, FELLOWSHIPS & GRANTS

- Phi Beta Kappa (Inducted as a Junior)
- Dean's List, Lake Forest College (2023 – Present)
- Presidential Scholarship, Lake Forest College (2023 – Present)
- Siemens Foundation Career Path Scholarship (2023 – Present)

PUBLICATIONS & MANUSCRIPTS

1. **Levenshteyn, J.**, Pandey, A., Chen, W., & Keten, S. "FEXS: A Novel Metric for Sequence-to-Property Explainability Evaluation in Machine Learning Models." *Under Review* (2026).

RESEARCH EXPERIENCE

Undergraduate Academic Year Researcher August 2026 – Present
Chen Lab & Keten Lab, Northwestern University Evanston, IL
Advisors: Dr. Wei Chen & Dr. Sinan Keten
Project: CAMEL – Causal Analysis of Materials Extracted from Literature

- Refining a completeness metric for LLM-extracted causal claims, measuring recall against human-annotated ground truth rather than correctness alone
- Developed a deduplication step that merges redundant claims extracted from the same paper, improving knowledge-graph quality and evaluation accuracy
- Automated the end-to-end pipeline, from PDF parsing through extraction, deduplication, and evaluation, minimizing human intervention at each stage
- Implemented batched LLM processing to scale extraction from single papers to large literature corpora
- Drafting a manuscript describing the CAMEL pipeline and evaluation framework

Undergraduate Summer Researcher June 2026 – August 2026
Chen Lab & Keten Lab, Northwestern University Evanston, IL
Advisors: Dr. Wei Chen & Dr. Sinan Keten
Project: CAMEL – Causal Analysis of Materials Extracted from Literature

- Extended CAMEL, an LLM-based pipeline that extracts causal claims from materials science literature into a Neo4j knowledge graph, to improve traceability and relationship granularity
- Designed a five-tier relation classification system (polarity, causal vs. correlational, functional form, equation, tabular fit) to extract structured causal claims
- Re-engineered the PDF-to-text stage, recovering 100% of equations and table data
- Constructed a hand-labeled ground-truth dataset of 200+ causal claims across three benchmark papers

REU+ Academic Year Researcher **August 2025 – May 2026**
Chen Lab & Keten Lab, Northwestern University *Evanston, IL*
Advisors: Dr. Wei Chen & Dr. Sinan Keten

- Designed and validated FEXS, a sign-invariant evaluation metric using rank-based importance masking and Spearman rank correlation to test whether feature attribution scores faithfully predict model behavior in sequence-to-property architectures
- Benchmarked FEXS across seven model–XAI combinations spanning three sequence-based architectures and a random-importance baseline, demonstrating it reliably distinguishes faithful from unfaithful importance scores
- Authored a first-author manuscript detailing the mathematical formulation, implementation, and benchmarking of FEXS, currently under review

MRSEC REU Summer Research Intern **June 2025 – August 2025**
Materials Research Science & Engineering Center, Northwestern University *Evanston, IL*
Advisors: Dr. Wei Chen & Dr. Sinan Keten

- Built an ensemble architecture that aggregated Multi-Motif COLOR models via weighted and unweighted averaging to improve amino acid-level importance predictions for sequence-to-property regression
- Designed a masking-based evaluation protocol — progressively unmasking amino acid sequences and tracking mean absolute error — to benchmark explainability across model variants against known indices (aliphatic index, GRAVY hydropathy) and an unknown-ground-truth case (instability index)
- Surfaced a bias in the masking evaluation approach when results proved inconsistent across property types, a finding that directly motivated my follow-on development of FEXS, a sign-invariant explainability metric

RELEVANT COURSEWORK

- **Mathematics & Foundations:** Tutorial Project: Tensors & Tensor Geometry (MATH-491), Calculus I, II, & III (Analytic Geometry III), Linear Algebra (MATH-231), Mathematical Probability (MATH-350), Abstract & Discrete Mathematics (MATH-230), Intro to Computational Math (CSCI-240).
- **Data Science & Computer Science:** Artificial Intelligence (CSCI-325), Computer Vision & Machine Learning (CSCI-450), Introduction to Database Systems (CSCI-327), Programming for Data Applications (CSCI-250), Computer Science I & II (Python & Java).
- **Physical & Chemical Sciences:** Physical Chemistry I & II (CHEM-320L/321L), Instrumental Analysis & Lab (CHEM-410L/410X), Advanced Organic Chemistry (CHEM-430), Biochemistry (BMB-300L), Inorganic Chemistry (CHEM-340L), Organic Chemistry I & II, General Physics I & II.

ACADEMIC PRESENTATIONS

- **Levenshteyn, J.**; Li, S.; Kumbhojkar, G.; Chen, W.; Keten, S. "Causal Analysis of Materials Extracted from Literature: Ontology-Based LLM Causal Claim Extraction," Poster, Glassman Symposium (October 2026) and Department of Chemistry Poster Session (October 2026), Lake Forest College, Lake Forest, IL.
- **Levenshteyn, J.** "Netflix Title Sentiment Analysis for IMDb Rating Prediction," Programming for Data Applications Presentation, Lake Forest College, Lake Forest, IL (December 2024).
- Adamik, A.; Chaoa, M.; **Levenshteyn, J.**; McKee, K.; Muchacho, K. "Neuroscience of Spinocerebellar Ataxias," Brain Awareness Symposium, Lake Forest College, Lake Forest, IL (December 2023).

TECHNICAL SKILLS & METHODS

Core AI & Data Science: Large Language Models (LLMs), Causal Extraction, Explainable AI (XAI / FEXS Metric), Ensemble Learning, Deep Neural Networks (CNNs, MLPs), Reinforcement Learning (RL), Sequence-to-Property Regression, Feature Engineering, Search Optimization Algorithms

Languages & Toolkits: Python (PyTorch, Scikit-learn, TensorFlow, NumPy, pandas), Java, C#, SQL, LaTeX, Git/GitHub, Jupyter Notebooks, VS Code, pgAdmin, TensorBoard, Stata, Jamovi, Neo4j

Laboratory & Instrumentation: NMR Spectroscopy, IR Spectroscopy, UV-Vis Spectroscopy, X-Ray Crystallography, X-Ray Fluorescence (XRF), Differential Scanning Calorimetry (DSC), Gas Chromatography, Fractional Distillation, Bomb Calorimetry, Flash Photolysis, Chemical Kinetics (conductometric and spectroscopic methods), Organic Synthesis & Purification

LEADERSHIP & SERVICE

Student Affiliates of the American Chemical Society (SAACS) **April 2025 – Present**
Co-President

- Secured funding through the Career Center and organized a chapter trip to the ACS National Meeting in Chicago
- Organize student research panels, student research poster sessions, and a faculty research session to connect students with research opportunities
- Collaborate with the Chicago Society of Younger Chemists and lead National Chemistry Week and Mole Day outreach activities
- Create a Diversity in Chemistry poster each semester, host bimonthly ChemisTea community events, and co-host an end-of-semester physical sciences picnic with the physics student organization

STREAM (Student Technical Research in ML & Stats) **August 2026 – Present**
Secretary

- Coordinate meetings and communications for a student research group focused on machine learning and statistics
- Support member-led research discussions and project presentations, maintaining records and shared resources for the group

First Generation Foresters **December 2024 – Present**
PR Director

- Joined the founding board and led marketing that grew the organization from zero members into an active student club on campus
- Promoted first-generation faculty panels, entrepreneurship panels, and community-building events
- Planned events such as the First-Gen Week, Career Spring event, and an end-of-year senior celebration

LANGUAGES

- Russian (fluent)
- Spanish (conversational)

REFERENCES

- **Sinan Keten, Ph.D.** – Associate Chair and Professor of Mechanical Engineering, Professor of Materials Science, Keten Lab, Northwestern University
- **Wei Chen, Ph.D.** – Chair and Professor of Mechanical Engineering, Professor of Materials Science, Chen Lab, Northwestern University
- **Sara Jamshidi, Ph.D.** – Associate Professor of Computer Science and Mathematics, Co-Chair of Artificial Intelligence, Lake Forest College
- **Dawn Wiser, Ph.D.** – Associate Professor of Chemistry, Lake Forest College